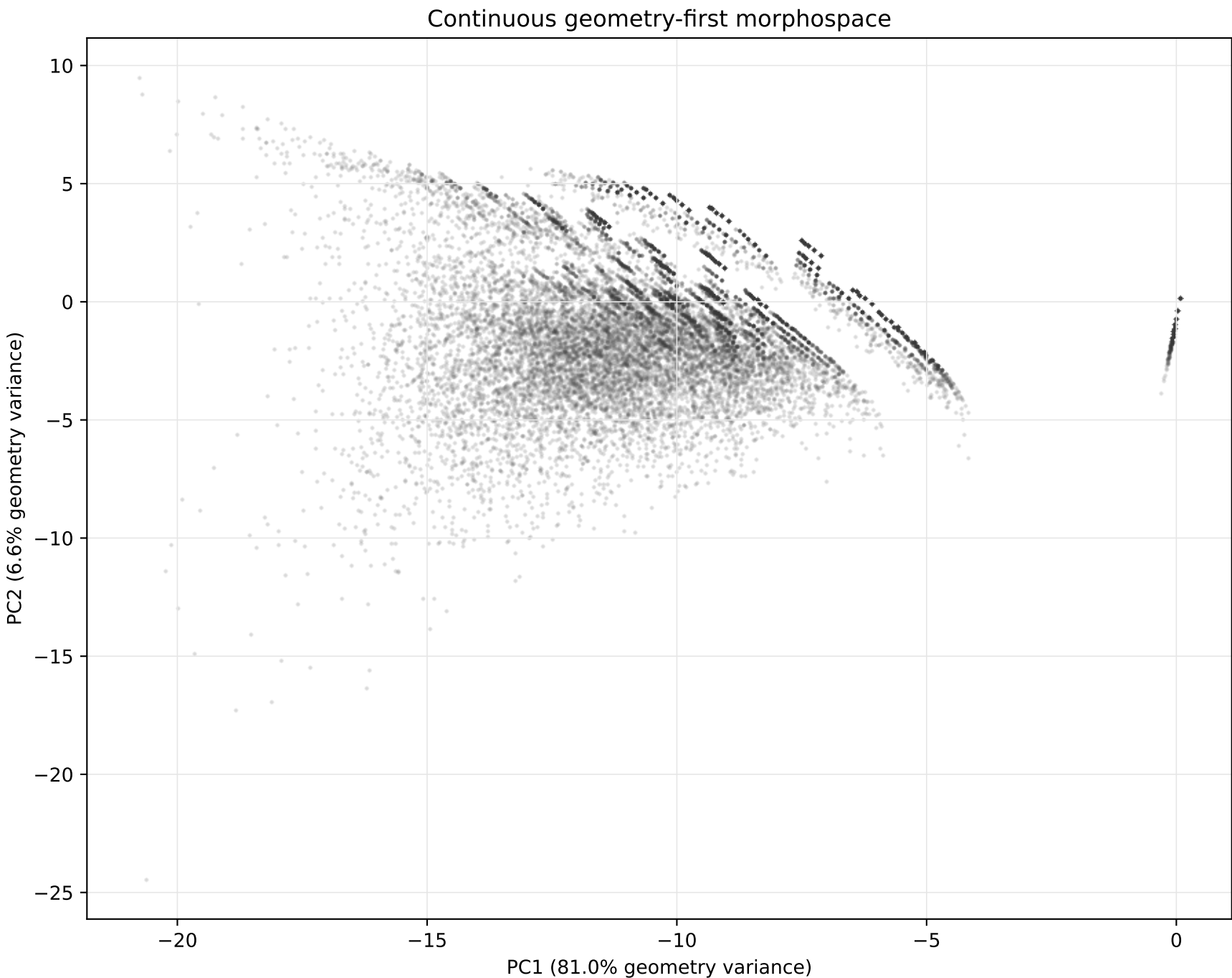


Fire VASE Developmental Morphospace

Geometry defines the morphospace; climate is attached afterward and evaluated as coupling.



Fires represented: 278,569
Climate-complete fires: 237,235
Geometry PC1-PC5 variance: 96.3%
Median duration: 1.0 days
Median final area: 0.43 km2

Interpretation: categories are labels placed inside the space, not the axes of the space. The first pass estimates coupling with linear information proxies, so low R2 should be read as weak linear recoverability rather than proof of no relationship.

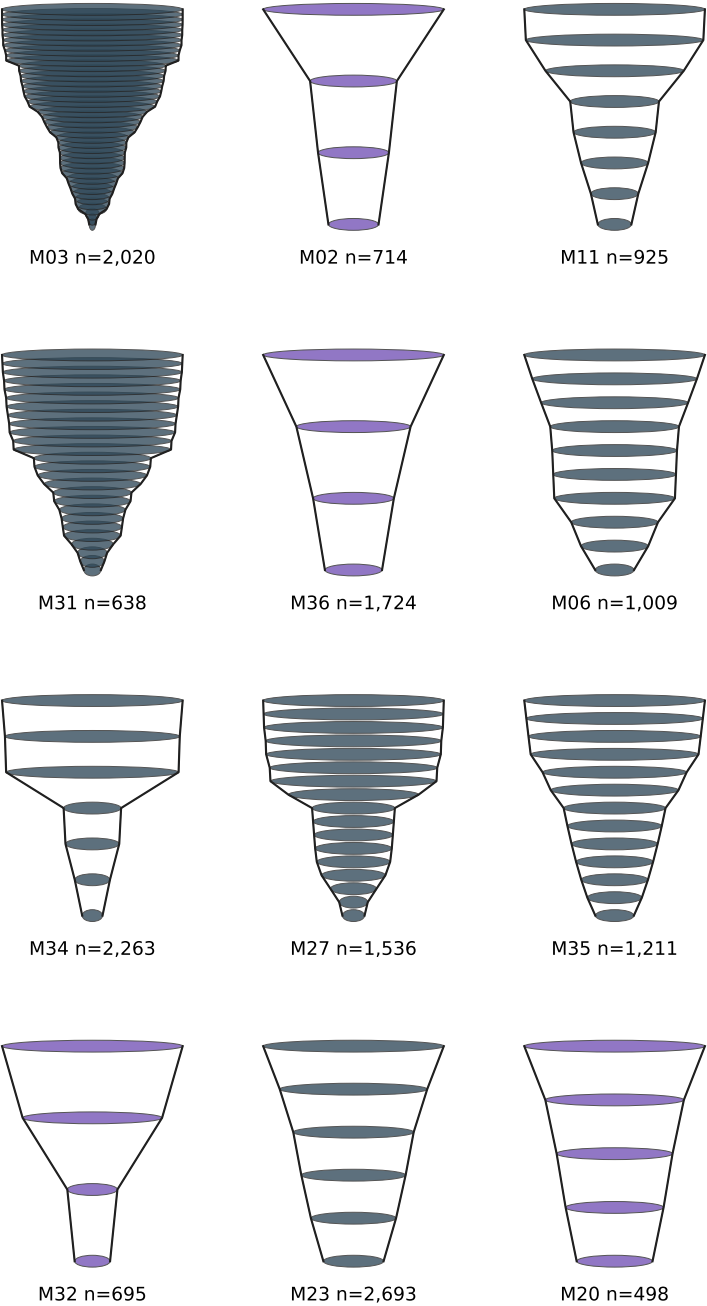
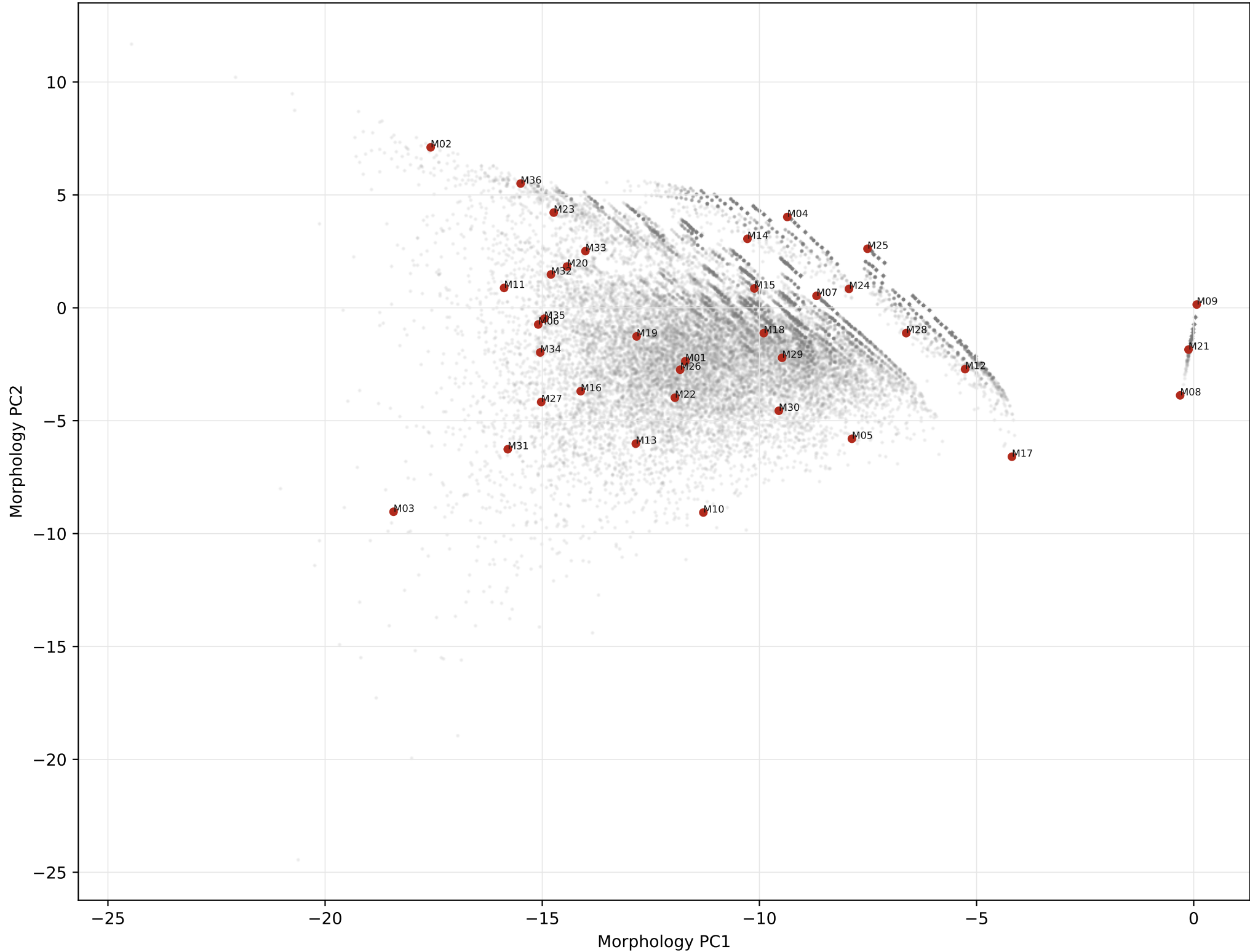
Mean directional coupling proxy

P(climate | morphology): R2=0.020

P(morphology | climate): R2=0.191

Figure 2: Morphospace With Representative Fire VASEs

Medoids are selected by farthest-point coverage in PC1-PC3. Counts are fires assigned to each nearest medoid.



What Dimensions Organize Wildfire Development?

Largest absolute geometry-feature loadings for the first three morphospace axes.

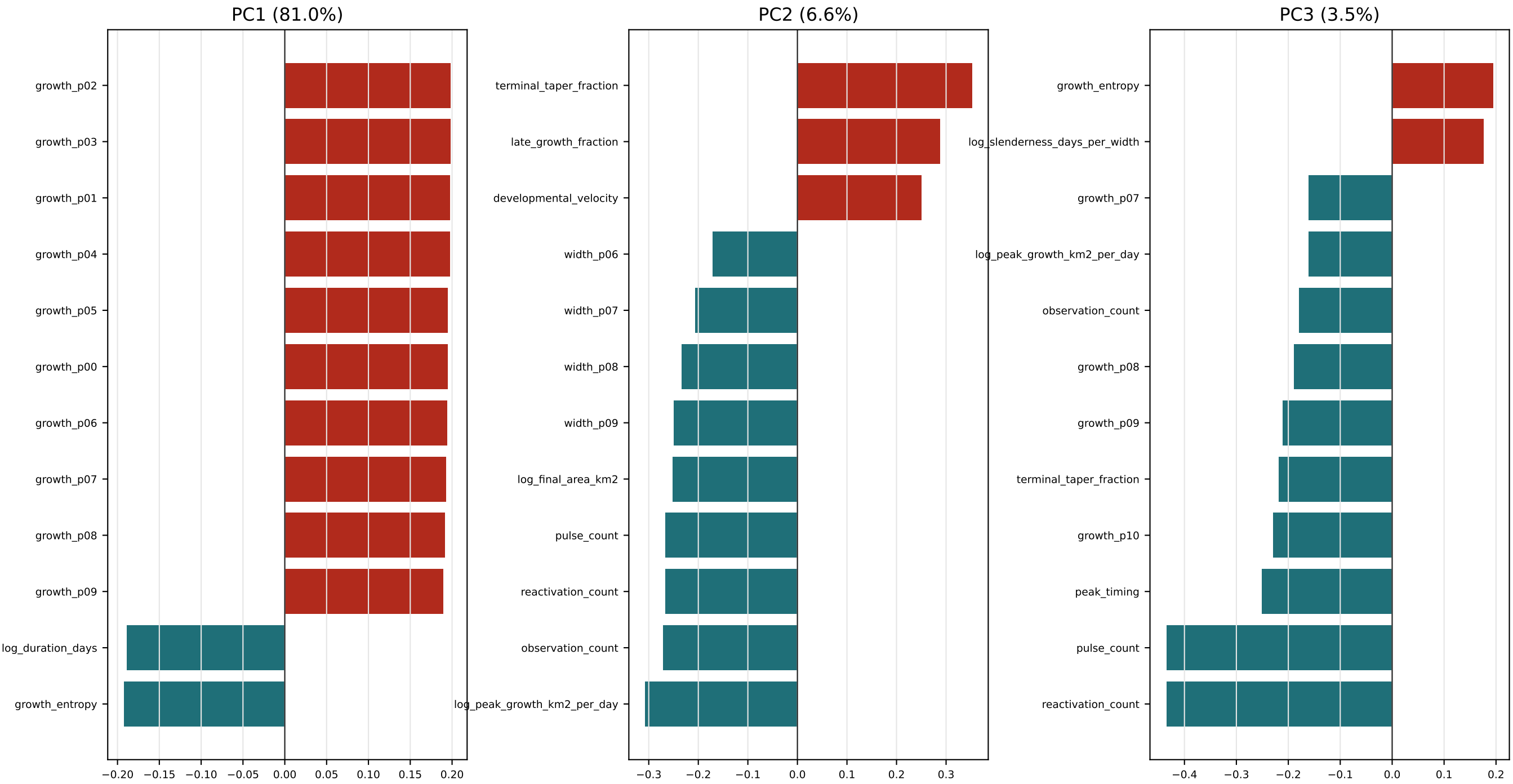
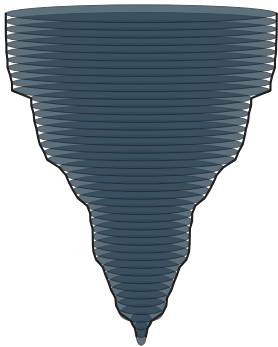
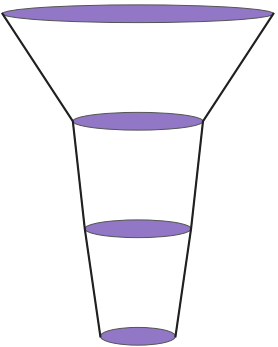


Figure 3: Morphological Field Guide, Page 1

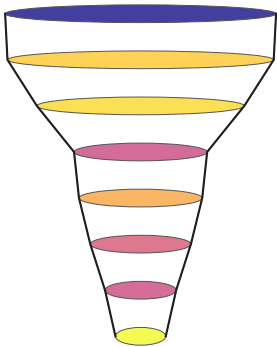
Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.



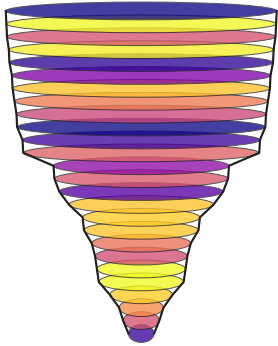
M03 fire 100734
multi-pulse complex | represents 2,020
area 405.9 km2 | duration 48.0 d | pulses 2
tmax nan C | VPD nan kPa | wind nan m/s
neighbors: 295915,418043,284572,369903,145582,80211



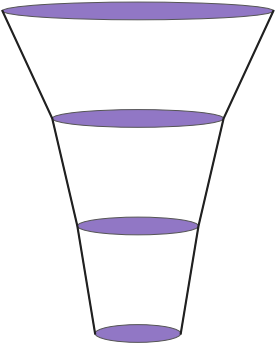
M02 fire 100344
late surge | represents 714
area 2.8 km2 | duration 5.0 d | pulses 1
tmax nan C | VPD nan kPa | wind nan m/s
neighbors: 63932,442313,294519,135719,336427,374552



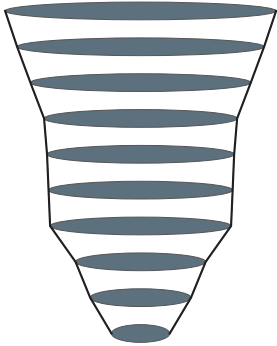
M11 fire 372443
multi-pulse complex | represents 925
area 6.2 km2 | duration 18.0 d | pulses 1
tmax 27.8 C | VPD 1.01 kPa | wind 3.7 m/s
neighbors: 110338,222386,212586,103932,429741,2734



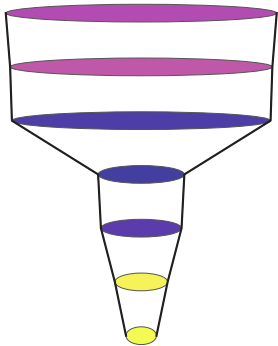
M31 fire 327634
multi-pulse complex | represents 638
area 24.0 km2 | duration 35.0 d | pulses 2
tmax 27.2 C | VPD 1.18 kPa | wind 3.7 m/s
neighbors: 314254,77068,275788,387579,319860,63259



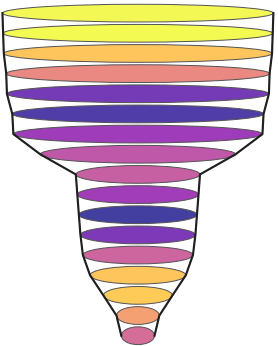
M36 fire 406385
late surge | represents 1,724
area 2.1 km2 | duration 9.0 d | pulses 1
tmax nan C | VPD nan kPa | wind nan m/s
neighbors: 414886,417011,149444,206223,437709,51749



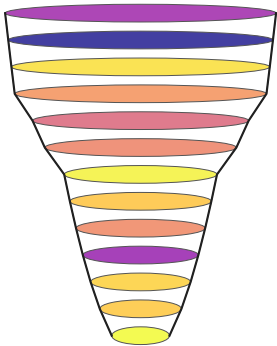
M06 fire 22848
multi-pulse complex | represents 1,009
area 14.0 km2 | duration 15.0 d | pulses 2
tmax nan C | VPD nan kPa | wind nan m/s
neighbors: 102807,159765,246306,416158,273935,3721



M34 fire 108371
multi-pulse complex | represents 2,263
area 17.0 km2 | duration 9.0 d | pulses 1
tmax 32.9 C | VPD 0.98 kPa | wind 6.3 m/s
neighbors: 114185,14591,422152,47650,142948,209683



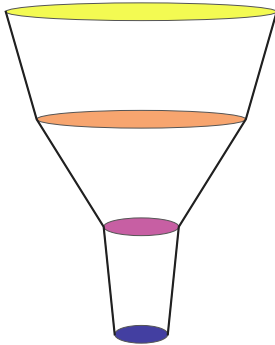
M27 fire 338158
multi-pulse complex | represents 1,536
area 42.9 km2 | duration 18.0 d | pulses 1
tmax 10.4 C | VPD 0.44 kPa | wind 4.9 m/s
neighbors: 186947,312647,327380,63734,182800,235569



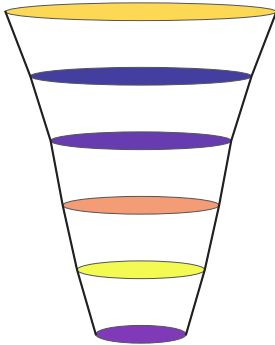
M35 fire 297991
multi-pulse complex | represents 1,211
area 4.7 km2 | duration 23.0 d | pulses 1
tmax 30.7 C | VPD 1.41 kPa | wind 3.8 m/s
neighbors: 373639,261902,191038,134458,328732,4913

Figure 3: Morphological Field Guide, Page 2

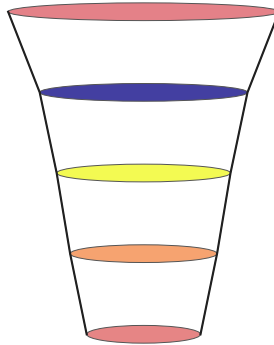
Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.



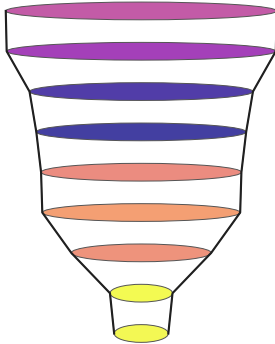
M32 fire 423200
late surge | represents 695
area 11.2 km2 | duration 5.0 d | pulses 1
tmax 18.1 C | VPD 0.68 kPa | wind 5.1 m/s
neighbors: 115473,6053,125993,228423,235216,379999



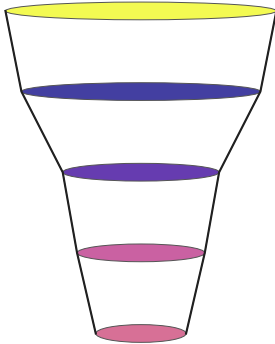
M23 fire 116797
multi-pulse complex | represents 2,693
area 1.9 km2 | duration 14.0 d | pulses 1
tmax 23.8 C | VPD 1.05 kPa | wind 3.9 m/s
neighbors: 28146,52136,111174,437657,57445,46988



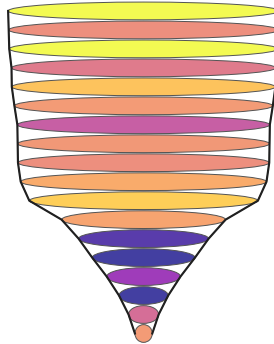
M20 fire 220581
late surge | represents 498
area 3.6 km2 | duration 6.0 d | pulses 2
tmax 25.4 C | VPD 1.51 kPa | wind 2.8 m/s
neighbors: 328305,400100,169709,353839,277949,2113



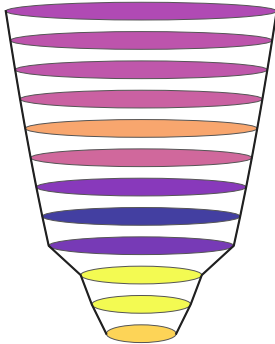
M16 fire 259789
multi-pulse complex | represents 1,153
area 16.1 km2 | duration 9.0 d | pulses 2
tmax 26.1 C | VPD 1.42 kPa | wind 2.0 m/s
neighbors: 321540,308539,272188,55292,168600,46560



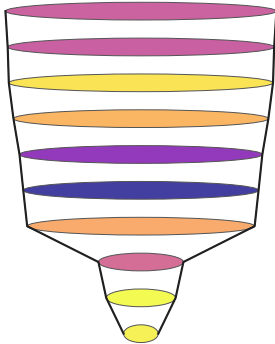
M33 fire 24803
late surge | represents 2,601
area 1.9 km2 | duration 14.0 d | pulses 1
tmax 31.7 C | VPD 0.97 kPa | wind 3.1 m/s
neighbors: 412547,390892,329872,239156,2258,98836



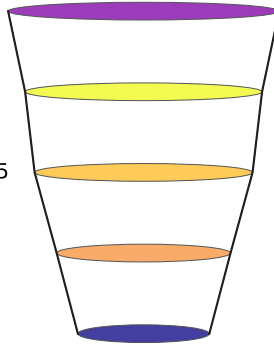
M13 fire 40025
multi-pulse complex | represents 788
area 52.2 km2 | duration 20.0 d | pulses 1
tmax 35.1 C | VPD 1.76 kPa | wind 4.8 m/s
neighbors: 275948,253531,20928,294559,42282,22147



M19 fire 9846
multi-pulse complex | represents 7,297
area 3.2 km2 | duration 22.0 d | pulses 1
tmax 25.1 C | VPD 0.85 kPa | wind 3.3 m/s
neighbors: 25853,46131,394795,300277,7969,149292



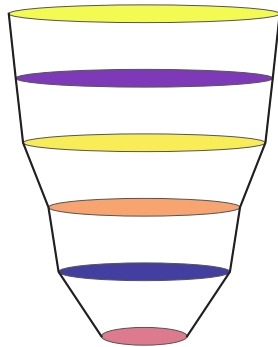
M22 fire 328682
multi-pulse complex | represents 3,082
area 13.1 km2 | duration 19.0 d | pulses 1
tmax 23.6 C | VPD 0.83 kPa | wind 4.1 m/s
neighbors: 328916,179082,208058,392042,123615,225895



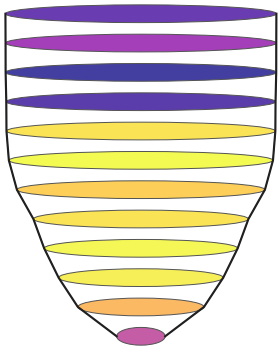
M26 fire 414357
multi-pulse complex | represents 738
area 3.6 km2 | duration 6.0 d | pulses 3
tmax 33.2 C | VPD 1.41 kPa | wind 3.9 m/s
neighbors: 113743,262821,222589,262063,114797,1269

Figure 3: Morphological Field Guide, Page 3

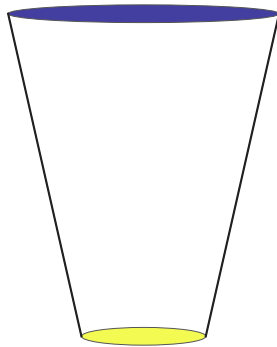
Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.



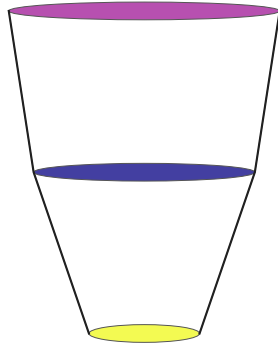
M01 fire 244766
multi-pulse complex | represents 3,264
area 2.1 km2 | duration 15.0 d | pulses 2
tmax 22.3 C | VPD 0.97 kPa | wind 3.9 m/s
neighbors: 48984,214573,200056,255093,286540,285243



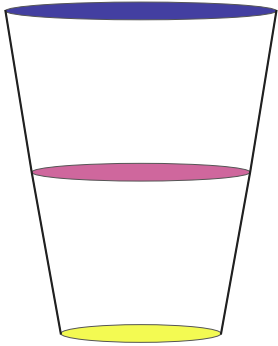
M10 fire 149229
multi-pulse complex | represents 897
area 254.6 km2 | duration 12.0 d | pulses 2
tmax 23.0 C | VPD 1.38 kPa | wind 4.0 m/s
neighbors: 6954,256555,147992,409527,327711,184820



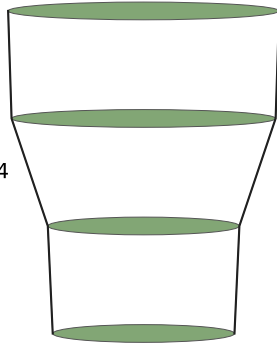
M14 fire 339492
late surge | represents 2,533
area 4.1 km2 | duration 2.0 d | pulses 1
tmax 23.6 C | VPD 1.19 kPa | wind 5.3 m/s
neighbors: 442160,379062,141477,440921,303838,5517



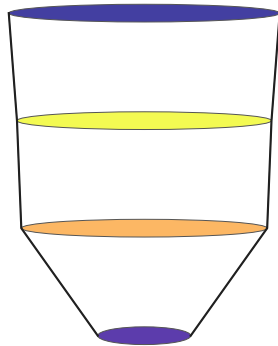
M15 fire 10074
compact steady | represents 10,626
area 1.3 km2 | duration 4.0 d | pulses 1
tmax 12.2 C | VPD 0.47 kPa | wind 3.2 m/s
neighbors: 239500,351669,338387,153223,121251,53341



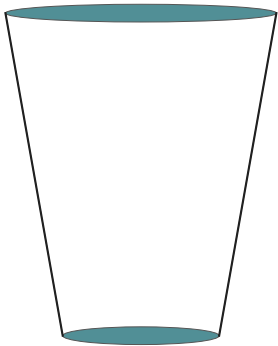
M18 fire 374851
compact steady | represents 1,891
area 4.3 km2 | duration 5.0 d | pulses 2
tmax 21.8 C | VPD 0.40 kPa | wind 3.0 m/s
neighbors: 174777,403428,248225,438710,229039,124204



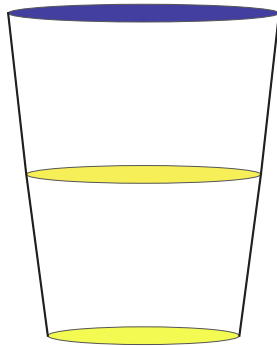
M30 fire 193258
compact steady | represents 1,845
area 4.3 km2 | duration 6.0 d | pulses 2
tmax nan C | VPD nan kPa | wind nan m/s
neighbors: 338872,181945,226560,113574,180753,8655



M29 fire 230310
front-loaded plateau | represents 9,735
area 3.6 km2 | duration 13.0 d | pulses 1
tmax 18.8 C | VPD 0.59 kPa | wind 5.7 m/s
neighbors: 432576,10073,200105,203167,12754,177191



M04 fire 100325
skinny persistent | represents 8,288
area 0.6 km2 | duration 6.0 d | pulses 1
tmax nan C | VPD nan kPa | wind nan m/s
neighbors: 10314,83713,266714,339472,396552,266858



M07 fire 263375
skinny persistent | represents 12,529
area 0.9 km2 | duration 11.0 d | pulses 1
tmax 26.7 C | VPD 1.03 kPa | wind 3.2 m/s
neighbors: 282569,397093,171874,376456,360423,3973

Figure 3: Morphological Field Guide, Page 4

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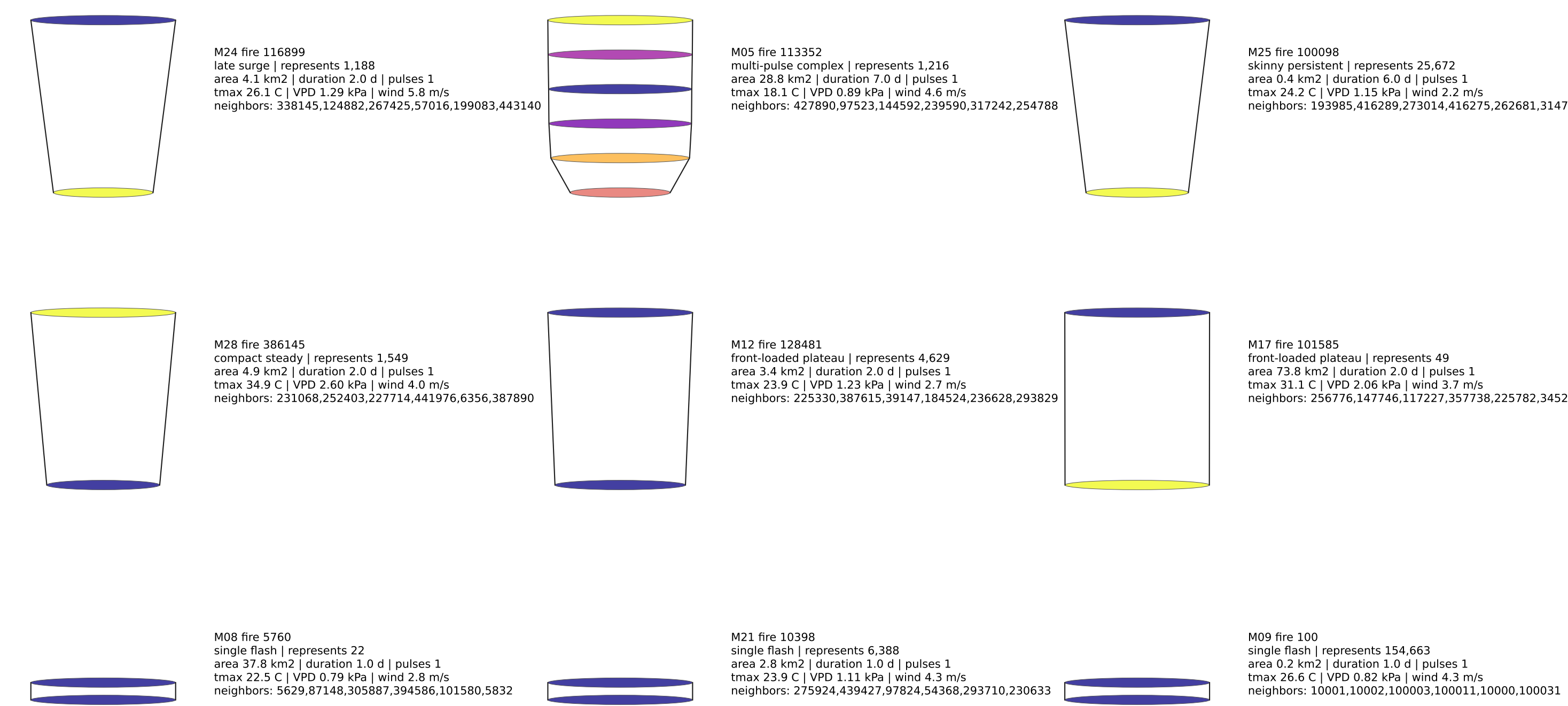


Figure 4: Climate Mapped Onto Morphospace

Climate is evaluated after geometry defines the morphospace.

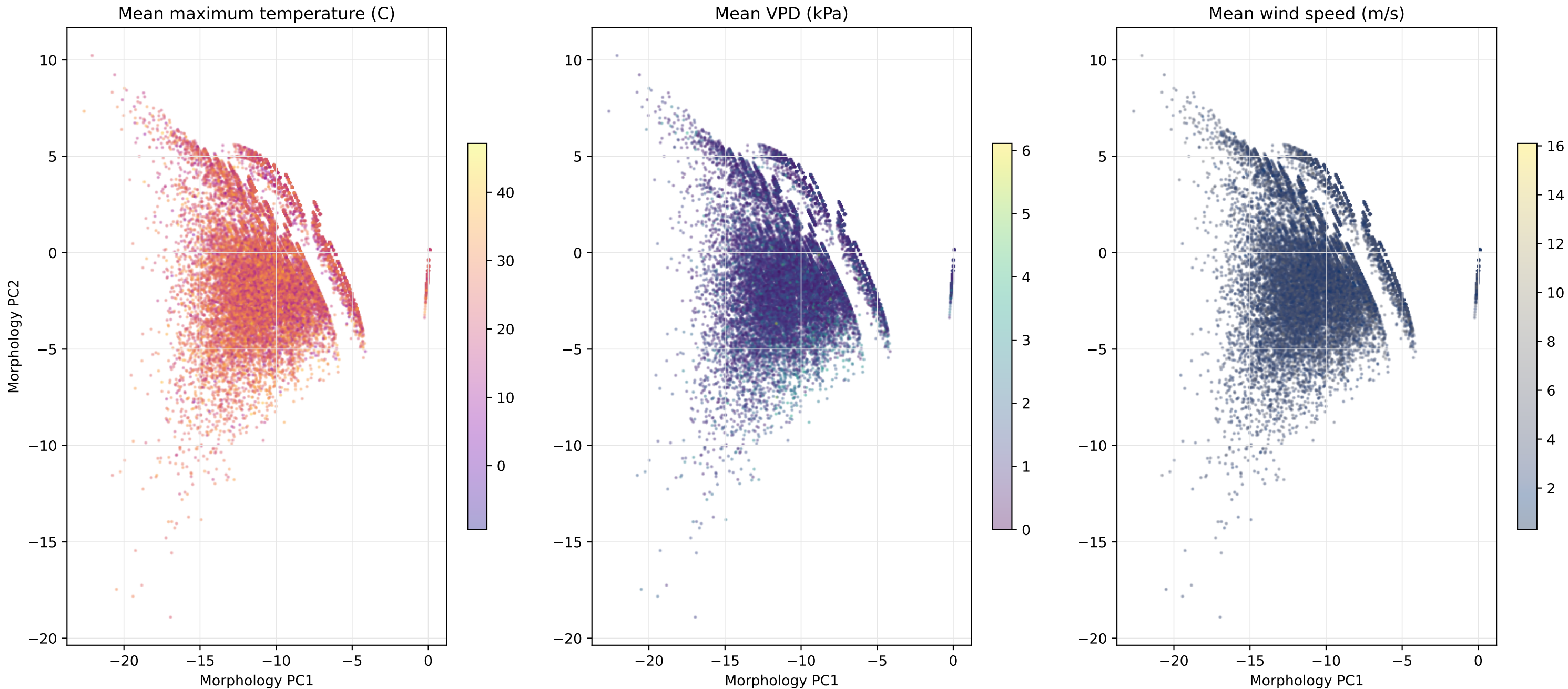


Figure 5: Matched Comparisons

Pairs estimate resilience: similar morphology under different climate, and similar climate with different morphology.

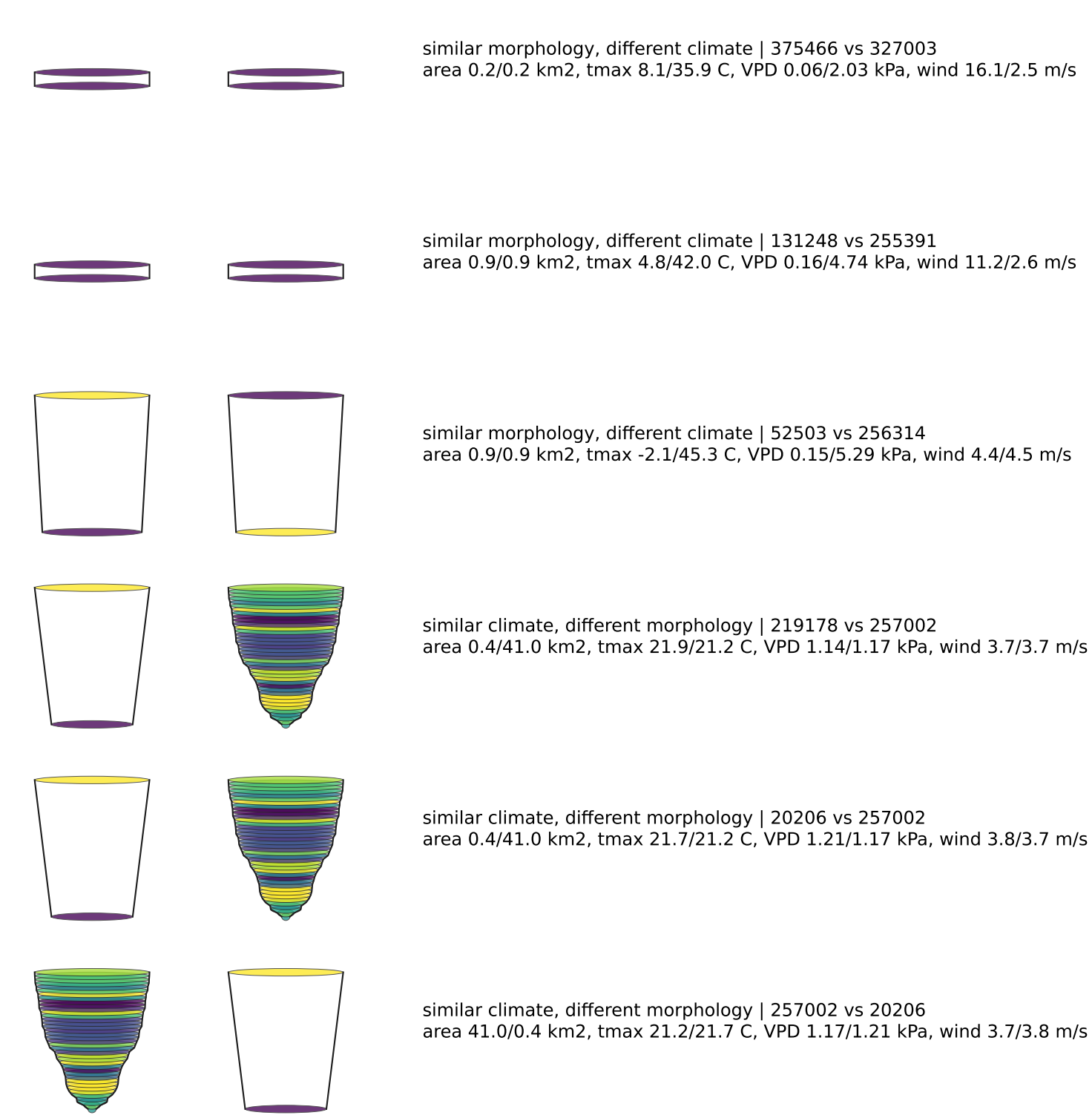


Figure 6: Developmental Control Profile

Predictive information proxies compare climate-only, geometry-only, and combined models through development.

